



Research paper

Prediction model of contact forces and IOP during digital palpation of porcine eyes

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ABSTRACT

Frequent intraocular pressure (IOP) measurements are desirable in the diagnosis and management of glaucoma. Most current tonometers utilize some form of corneal deformation to estimate the IOP, since trans-scleral tonometry suffers from loss of sensitivity. Tran-scleral and trans-palpebral tonometry, however, offer a pathway towards a non-invasive home tonometry. This article presents a mathematical model capturing the relationship between the IOP and the displacements imposed onto the sclera by externally applied forces. Similar to manual digital palpation tonometry, trans-scleral mechanical palpation makes use of two force probes that are advanced in a specific order and distance. Data from the applied forces and displacements, along with concurrent measurements of IOP is used to produce a phenomenological mathematical model. The experiments were carried out on enucleated porcine eyes.

Two models are presented. Model 1 predicts IOP vs forces and displacements, while Model 2 predicts the baseline IOP (prior to applying the forces) as a function of the measured forces and displacements. The proposed models result in IOP errors of 1.65 mmHg and 0.82 mmHg, respectively. Model parameters were extracted using least-squares-based system identification methods. The results show that the proposed models can be used to estimate the baseline IOP with accuracy of ± 1 mmHg over a pressure range of 10–35 mmHg, solely from measurement of tactile forces and displacements.

1. Introduction

Analysis of the bio-mechanical properties of the eye and its tissues is important for diagnosis and management of glaucoma, retinal vein occlusion (RVO) and even neurological disorders of the central nervous system associated in increased intracranial pressure (ICP) (Song et al., 2019; Jabbehdari et al., 2022). Therefore, measurement and prediction of IOP is important non-invasive method for the diagnosis of multiple diseases (Price et al., 2020; Lashutka et al., 2004; Nabeta et al., 2014). Among these, glaucoma is the best known disease that requires frequent measurement of eye pressure (Jonas et al., 2017; Matlach et al., 2019; Investigators et al., 2000). It is one of the leading causes of irreversible blindness worldwide, with an estimated global prevalence of 3.54% (Tham et al., 2014; Quigley and Broman, 2006). Traditional criterion for the presence of glaucoma is an IOP higher than 21 mmHg (Sommer et al., 1991).

Management of glaucoma includes use of IOP reducing drugs (Lusthaus and Goldberg, 2019), or placement of a drainage shunt in order to control the eye pressure. Because of diurnal variations of IOP, as high as 3–5 mmHg, its frequent measurements at home are desirable,

but difficult to implement. Most current techniques require office visit, use of toxic anaesthetics, and a sterile instrument to aplanate the cornea. Similar to Goldmann tonometry, many of these alternative tonometers are affected by central corneal thickness, curvature, or biomechanical properties (Liu and Roberts, 2005). An alternative to these methods is a mechanical palpation of the eye through the eye lid, which was first reported more than 3000 years ago (Stamper, 2011), but is deemed inaccurate today. The ability to carry out through-the-eyelid tonometry is attractive, as it could lead to the development of new sensors for home use.

This article reports on an experimental study evaluating the correlation between contact forces, displacements, and IOP during mechanical palpation of porcine eyes through the sclera. The approach emulates the digital palpation tonometry, where two probes (fingers) are pressed against the eye in alternating pattern, in order to perceive its stiffness and hence IOP. This paper describes the analysis of an instrumented version of this test applied directly to the sclera (trans-scleral tonometry). In the test the displacement, the resulting forces, and IOP are recorded simultaneously. As demonstrated here, during such test, the

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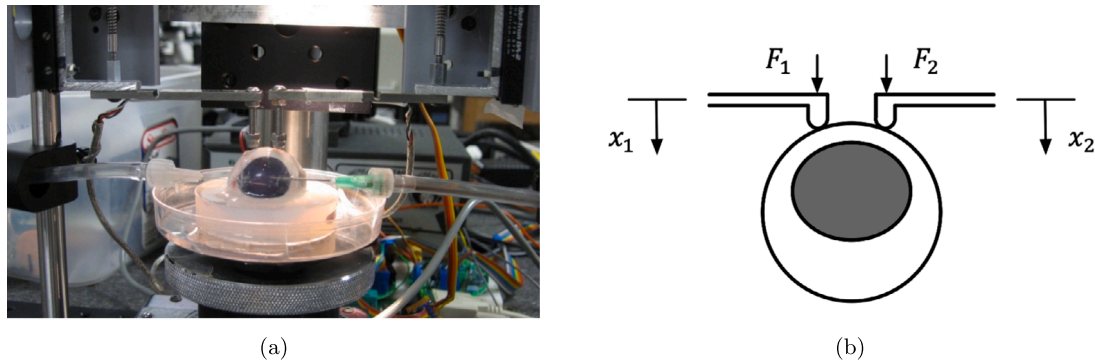


Fig. 1. (a) Palpation experimentation platform (b) Schematic diagram of porcine eyeball and two indenters.

IOP increases significantly. The data from the reported experiments are used to create a mathematical model that can predict the changes of IOP and palpation forces using different starting levels of IOP. It is also demonstrated that once calibrated, the model can predict the IOP changes from measurements of the forces and displacements of the palpating probes. Furthermore, it is demonstrated that the initial (baseline) IOP can be estimated from sole measurements of forces and displacements, hence providing a possible technique for estimating IOP using a calibrated model. While the investigation did not include the eye-lid, its role can be minimized by appropriately sizing the force probes as explained in the conclusions section of this work.

2. Method and apparatus

2.1. Experimental apparatus

The goal of this experiment is to mimic palpation and collect data to “describe” the feelings on fingers during a physical examination. It was intended to have an experiment that could apply forces on the eyeball in a variety of time-varying ways and have control over a reasonable number of distinct displacements of force sensors. Two indenters as the main part of the palpation experimental platform are shown in Fig. 1(a). Computer-based position control system was used to construct motorized linear translation stages. The delivered resolution is 0.0254 mm per step by linear actuator, which is driven by stepper motor. The test eyeball is placed on a plate. For the initial positioning, three manual linear stages and one manual rotational stage were used.

The study employed enucleated porcine eyeballs from First Vision Tech in Mesquite, Texas. The pigs that were sacrificed ranged in age from two to five. The eyeballs were enucleated, put in a sealed plastic bag with saline solution. The eyeballs were analyzed 3–4 days after the death in a laboratory refrigerator maintained at 4 °C. The experiments were conducted in a 22 °C laboratory environment. The artificial eye socket was created using a 30 g/l agarose solution in water. The test eyeball was placed in a 55 mm-diameter petri dish and almost half of the globe was embedded in the socket.

Manometry is the most precise approach to measure the pressure inside the eye and is a common laboratory technique for evaluating changes in IOP over time. A hollow needle that is surgically inserted into the anterior chamber also makes manometry the most invasive approach and usually used for enucleation or intraocular surgery and as reference for other IOP measurement methods (Knietedt et al., 2008). Due to the high precision, a manometer as described in Hallberg and Print & Media (2006) is used for regulation and measurement of IOP. A plastic syringe filled with the 1% saline solution, and a three-way valve were linked via PVC tubing. One of the three-way valve's outputs was connected to a commercial pressure sensor, while the other outlet was connected to a thin-walled syringe needle. The needle was introduced into the vitreous humor during intravitreal pressurization, and it was then sealed with super glue. The transcorneal approach

required injection via the cornea. According to this procedure, the needle is placed into the aqueous chamber's core after passing through the limbus.

With this experimental set-up, the motorized palpations are able to carry out at different starting IOP values. Meanwhile, the data acquisition, which includes two forces, two displacements and changing IOP, are fulfilled in real-time.

2.2. Eye indentation protocol

The following test protocol was used to extract the parameters of the model (1)–(3). With reference to Fig. 1(b), Probe 1(left) and Probe 2(right) were advanced simultaneously using constant rate of 0.4 mm/sec to 2 mm. Subsequently, Probe 1 continued for an additional 0.5 mm while Probe 2 kept still, and was then retracted back to a position of 2 mm. Finally both probes were retracted concurrently from 2 mm to 0 mm. It is assumed that during manual palpation, the stiffness is determined when both probes (fingers) are in good contact with the eye, i.e. there is a certain level of pressure. To emulate this, only data where Probe 2 is still, displaced by 2 mm, while Probe 1 is advancing to reach 2.5 mm, is used. This corresponds to the final period from 5-th to the 6-th seconds in Fig. 2.

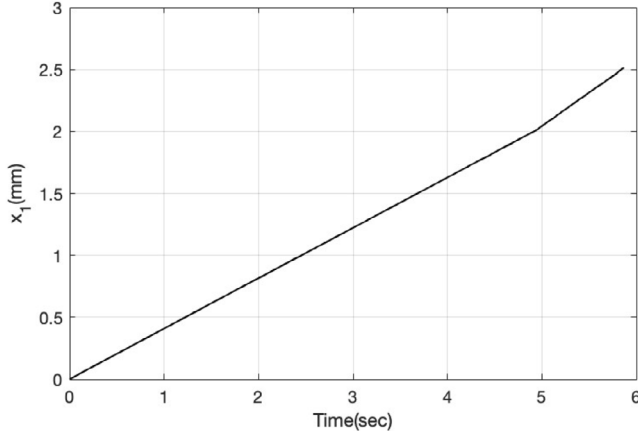
Prior to each experiment, the starting IOP, P_0 , was set to a pre-determined value in increments of 5 mm Hg in the range of 10, 15, ..., 35 mm Hg. The pressure was adjusted using a manometric column with a 3-way valve. After the pre-pressurization, the valve was closed to emulate the intact eye conditions. The resulting forces and IOP were recorded as shown in Figs. 3 and 4, respectively.

2.3. Model

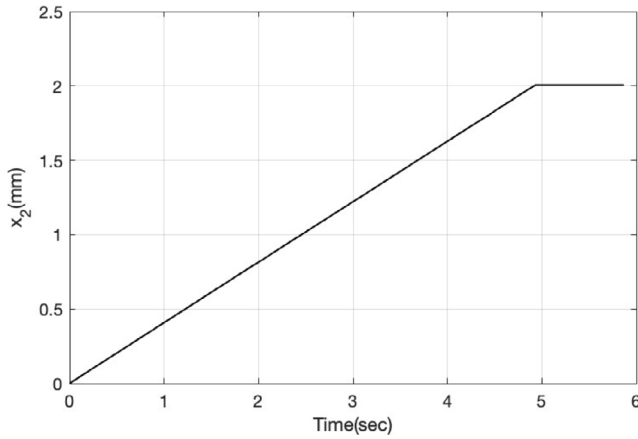
System identification techniques, guided by physical considerations were used to develop a phenomenological model describing the eye palpation. Fig. 1(b) shows a schematic of the two problems measuring forces F_1 and F_2 , while undergoing displacements x_1 and x_2 , respectively. The probes were placed symmetrically with respect to the sagittal plane of the eye, assisting in the development of the model. Our prior studies of the deformation and resulting IOP have shown that the force vs displacement, and IOP vs displacement have a progressive characteristic due to the non-linear behavior of the cornea-scleral shell and the incompressibility of the aqueous humor and vitreous (Polyvás et al., 2013a).

The model was build in two parts. First a pressure vs force and displacement was developed and subsequently a force vs displacement and pressure was added. The pressure model shown in (1) has two contributions. The first term represents the average effect of the applied forces on the IOP change, while the second accounts for the additional stretching the sclera when the two displacements are not equal (asymmetric palpation). Stretch of the corneo-scleral shell leads to higher IOP due to incompressibility of its contents.

$$P - P_0 = \alpha(F_1^m + F_2^m) + \beta|x_1 - x_2| \quad (1)$$



(a) Probe 1



(b) Probe 2

Fig. 2. Displacement of each probe.

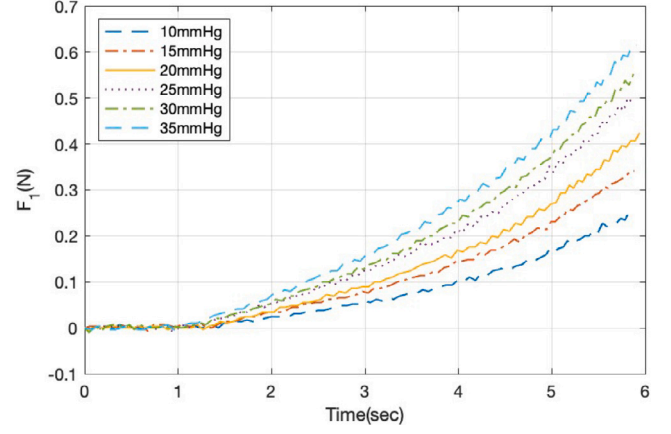
Prior analysis of effect of applying mechanical compression of the eye showed that IOP increases progressively as a function of the induced deformation (Polyvás et al., 2013b). This is due to stiffening of the corneo-scleral shell due to collagen fiber alignment in the direction of applied strain (Meek and Boote, 2004), as well as geometric effects associated with change of the direction of the tensile forces due to indentation. The increased IOP in turn, leads to progressive increase of the forces required to maintain the induced deformation. Accounting for these effects, we propose a force vs displacement and IOP model, that contains power series of the displacement that includes cubic terms when combined with (1)

$$(1 - \zeta)F_1 + \zeta F_2 = \gamma_1 P x_1 + \gamma_2 P_0 x_1 + \gamma_3 P x_2 + \gamma_4 P_0 x_2 + \delta_1 P x_1 x_2 + \delta_2 P_0 x_1 x_2 + \epsilon_1 P x_1^2 + \epsilon_2 P_0 x_1^2 + \epsilon_3 P x_2^2 + \epsilon_4 P_0 x_2^2 \quad (2)$$

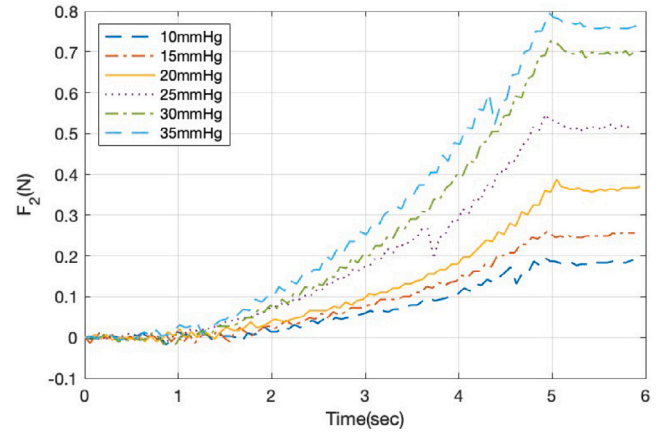
A symmetric model for the second probe is also used

$$(1 - \zeta)F_2 + \zeta F_1 = \gamma_1 P x_2 + \gamma_2 P_0 x_2 + \gamma_3 P x_1 + \gamma_4 P_0 x_1 + \delta_1 P x_1 x_2 + \delta_2 P_0 x_1 x_2 + \epsilon_1 P x_2^2 + \epsilon_2 P_0 x_2^2 + \epsilon_3 P x_1^2 + \epsilon_4 P_0 x_1^2 \quad (3)$$

The model (1)–(3) contains a total of 14 parameters. A least-squares technique (Keesman and Keesman, 2011) was used to estimate the values of the 14 parameters based on minimization of the errors between the measured and predicted values of the output forces, F_i , and the IOP,



(a) Probe 1



(b) Probe 2

Fig. 3. Detected force of each probe.

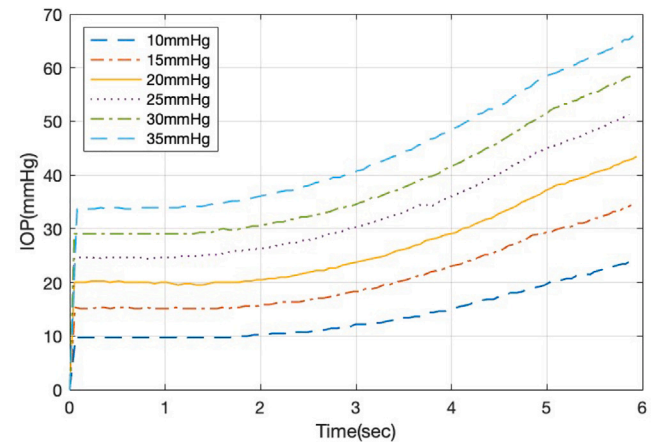


Fig. 4. Changing of IOP during experimental process.

P . Eq. (1) is non-linear in its parameter m , hence it required an iterative pseudo-least-squares technique. Briefly, the procedure represents the input-output relationship (1) as a non-linear function

$$y(i, \theta) = f(i, \theta) + e(i), \quad (4)$$

Table 1
Parameters for Model(1)–(3).

Parameters	Values	Units
m	0.6733	lnmmHg/lnN
α	17.3319	mmHg
β	6.8451	mmHg/mm
γ_1	0.0051	N/(mm·mmHg)
γ_2	−0.0108	N/(mm·mmHg)
γ_3	0.0051	N/(mm·mmHg)
γ_4	−0.0108	N/(mm·mmHg)
δ_1	−0.0833	N/(mm ² ·mmHg)
δ_2	0.1738	N/(mm ² ·mmHg)
ϵ_1	0.0387	N/(mm ² ·mmHg)
ϵ_2	−0.0787	N/(mm ² ·mmHg)
ϵ_3	0.0387	N/(mm ² ·mmHg)
ϵ_4	−0.0787	N/(mm ² ·mmHg)
ζ	0.5000	–

where i is the time stamp or measurement instance and $\theta = [\theta_1, \dots, \theta_p]^T$ is a parameter vector containing p parameters, and $e(i)$ are measurement errors. Taking the Jacobian of (4), one obtains the sensitivity matrix, $X(\theta)$ as

$$X(\theta) = \begin{pmatrix} \frac{\partial y(1;\theta)}{\partial \theta_1} & \frac{\partial y(1;\theta)}{\partial \theta_2} & \dots & \frac{\partial y(1;\theta)}{\partial \theta_p} \\ \frac{\partial y(2;\theta)}{\partial \theta_1} & \frac{\partial y(2;\theta)}{\partial \theta_2} & \dots & \frac{\partial y(2;\theta)}{\partial \theta_p} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y(N;\theta)}{\partial \theta_1} & \frac{\partial y(N;\theta)}{\partial \theta_2} & \dots & \frac{\partial y(N;\theta)}{\partial \theta_p} \end{pmatrix}, \quad (5)$$

where N is the number of measurement points. Using a step size $\sigma^{(j)} = 1$ and an initial guess $\hat{\theta}^{(0)} = [10, \dots, 10]$, a Gauss–Newton iteration is carried out in order to obtain an estimate of the parameters $\hat{\theta}$

$$\hat{\theta}^{(j+1)} = \hat{\theta}^{(j)} + \sigma^{(j)} X(\hat{\theta}^{(j)})^+ (y - f(i, \hat{\theta}^{(j)})). \quad (6)$$

Eqs. (2)–(3) are linear in their parameters, hence allowing single-step least-squares estimation. In the case of a linear parameter estimation problem, the input of the system is described as

$$y(i) = \Phi_1(i)\theta_1 + \dots + \Phi_p(i)\theta_p + e(i), \quad (7)$$

where $\Phi_j(i)$ are known (measured) regressor functions, e.g. $x_1(i), x_2(i), P(i), \dots$. In this case the sensitivity matrix does not depend on the parameters

$$X = \begin{pmatrix} \Phi_1(1) & \Phi_2(1) & \dots & \Phi_p(1) \\ \Phi_1(2) & \Phi_2(2) & \dots & \Phi_p(2) \\ \vdots & \vdots & \ddots & \vdots \\ \Phi_1(N) & \Phi_2(N) & \dots & \Phi_p(N) \end{pmatrix}$$

The parameter estimate is then computed through pseudo-inverse

$$\hat{\theta} = (X^T X)^{-1} X^T y, \quad (8)$$

where $y = [y(1), \dots, y(N)]^T$. Values of each parameter are shown in Table 1.

Since some of the practical applications of this model involve estimation of the initial pressure, P_0 , we have carried out a numerical experiment of combining Eqs. (1)–(3) into a single equation, by substituting P from (1) into (2) and (3). Due to symmetry of the geometry of the eye and locations of probes, (2) and (3) have the same coefficients, and hence only a single equation is obtained in the form $f(x_1, x_2, F_1, F_2, P_0; \mathbf{c}) = 0$, given by

$$\begin{aligned} f = & c_2(F_1^{c_1} + F_2^{c_1})x_1 + c_3|x_1 - x_2|x_1 + c_4P_0x_1 \\ & + c_5(F_1^{c_1} + F_2^{c_1})x_2 + c_6|x_1 - x_2|x_2 + c_7P_0x_2 \\ & + c_8(F_1^{c_1} + F_2^{c_1})x_1x_2 + c_9|x_1 - x_2|x_1x_2 + c_{10}P_0x_1x_2 \end{aligned}$$

Table 2
Parameters for Model(9).

Parameters	Values	Units
c_1	−9.9083	lnmmHg/lnN
c_2	8.8357e−05	N/mm
c_3	5.8911	N/mm ²
c_4	−0.0740	N/(mm·mmHg)
c_5	−8.4746e−05	N/mm
c_6	−6.2634	N/mm ²
c_7	0.0544	N/(mm·mmHg)
c_8	−4.4027e−05	N/mm ²
c_9	−1.5259	N/mm ³
c_{10}	0.0391	N/(mm ² ·mmHg)
c_{11}	−2.5856e−09	N/mm ²
c_{12}	−0.5633	N/mm ³
c_{13}	−1.5532e−04	N/(mm ² ·mmHg)
c_{14}	4.2230e−05	N/mm ²
c_{15}	2.2562	N/mm ³
c_{16}	−0.0250	N/(mm ² ·mmHg)
c_{17}	0.4159	–

$$\begin{aligned} & c_{11}(F_1^{c_1} + F_2^{c_1})x_1^2 + c_{12}|x_1 - x_2|x_1^2 + c_{13}P_0x_1^2 \\ & c_{14}(F_1^{c_1} + F_2^{c_1})x_2^2 + c_{15}|x_1 - x_2|x_2^2 + c_{16}P_0x_2^2 \\ & + (c_{17} - 1)F_1 - c_{17}F_2 = 0 \end{aligned} \quad (9)$$

The parameter vector $\mathbf{c} = (c_1, \dots, c_{17})$ in the above equation contains 17 parameters, due to the combinations of various products of the parameters in Table 1. Applying least-squares method to (9) with the data from Figs. 2–3 results in the parameters given in Table 2.

Once the parameters $\mathbf{c}$ are known, one can use (9) to compute P_0 from force and displacement data obtained under unknown P_0 .

3. Model evaluation

The validity of model (1)–(3) and the derived model (9) was tested over the entire range of test data. Since (9) contains more parameters than the model (1)–(3) (17 vs 14), it is expected that (9) will provide a better fit to the test data. It is also more useful model, since the pressure P provided by (1) cannot be measured in an in-vivo experiment. Therefore model (9) was evaluated first. To this end, the initial pressure P_0 was varied incrementally from 10 to 35 mmHg in steps of 5 mmHg. Data from Table 2 was used to compute the pressure P_0 from (9). Only force and displacement data were used, i.e. emulating a scenario where a tactile tonometer is applied without explicit knowledge of P or P_0 . The resulting plot in Fig. 5 shows the predicted P_0 vs the experimentally set value. For clarity, the units of the x -axis is virtual time, i.e. assembled from several consecutive tests representing the six levels of P_0 . The largest deviation is 0.85 mmHg occurs at $P_0 = 25$ mmHg.

In a similar fashion, the model (1)–(3) was evaluated. The resulting plot is shown in Fig. 6. As expected, the predicted P_0 from (1)–(3) has a larger maximum error 1.65 mmHg occurring at 30 mmHg.

Based on the above results, model (9) successfully keeps the errors of the estimated P_0 within ± 1 mmHg. However, comparison between parameters in Tables 1 and 2 show that some derived parameter values change. For example, $\zeta = 0.5$ changes to its corresponding parameter $c_{17} = 0.4159$ in Table 2, suggesting that the assumption of perfect symmetry does not hold.

4. Conclusions

This article presented a trans-scleral mechanical palpation study of enucleated porcine eyes with the goal of obtaining mathematical

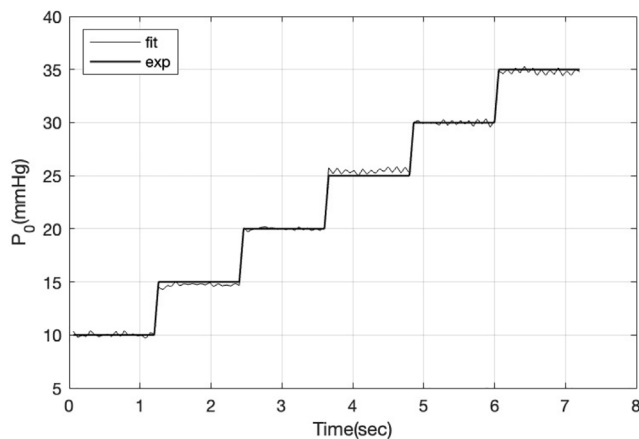


Fig. 5. Actual initial IOP vs estimated initial IOP from model (9).

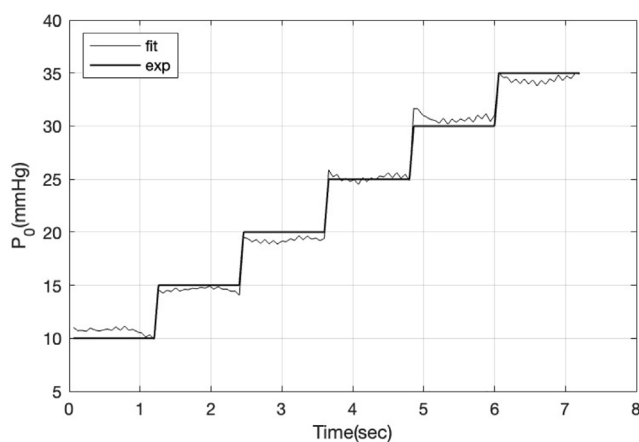


Fig. 6. Actual initial IOP vs estimated initial IOP from model (1)–(3).

models describing the dependence of the tactile forces, displacements, and intraocular pressure. Two models were evaluated. Model 1 is comprised of Eqs. (1)–(3) and Model 2 by Eq. (9). Both models estimated the initial pressure P_0 with accuracy below 2 mmHg, which is the industry standard for commercial tonometers. Model 2 (9) has higher accuracy due to larger number of parameters. Furthermore it provides estimate of P_0 solely based on measurements of forces and displacements. It exhibits a sufficiently high accuracy of 0.85 mmHg in the entire range of 10–35 mmHg IOP. Most commercial tonometers are certified to be within ± 2 mmHg to ± 3 mmHg against Goldmann application tonometry (Yilmaz et al., 2014; Subramaniam et al., 2021) and typically suffer from larger errors for low ($P_0 < 11$) high ($P_0 > 21$) (Wang et al., 2018). It is likely that Model 2 can be used in the development of non-invasive tactile tonometers.

Model 1, while somewhat less accurate, has the advantage of also predicting the IOP, P as a function of the applied forces and displacements via Eq. (1). Its utility is in applications where one aims to generate a desired IOP via application of external forces to the eye, for example as a non-invasive alternative to the Priscoll provocative test (Primrose, 1962).

In the present study, the existence of an eyelid was ignored, the probes are assumed to be set symmetrically and advanced rectilinearly towards the sclera using a motorized system. The eyelid contains skin, several structures such as muscle fibers and glandular tissue (tarsal plate) made up of dense connective tissue and Meibomian glands. The stiffness of the eye lid (its skin) is proportional to the area of the contact probes, while the stiffness of the scleral shell contains membrane forces proportional to the perimeter of the probe. Using larger probes,

could preferentially probe the stiffness of the scleral shell, however most likely a practical IOP measurement system will require subject-specific calibration. Such investigation could utilize similar approach as presented here, but is beyond the scope of the present work.

CRedit authorship contribution statement

Qiuchen Zhang: Validation, Formal analysis. **Eniko T. Enikov:** Writing – original draft, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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